

TEJAS NAIK J

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Professional Summary

Final-year Computer Science Engineering undergraduate (PES University) with hands-on research and full-stack development experience. Published co-author on two peer-reviewed conference papers covering AR-based navigation systems and attention-based hallucination detection for Retrieval-Augmented Generation (RAG) pipelines. Proficient in Python, machine learning, LLM-based agentic systems, and scalable backend development using Docker, Kubernetes, and Redis. Seeking Software Engineer / ML Engineer roles to apply research-driven problem-solving to production systems.

Education

B.Tech in Computer Science and Engineering

PES University, Bengaluru | CGPA: 7.89

2022 – 2026 (Expected)

Pre-University (PCMC)

Vidya Mandir Independent PU College, Bengaluru

2020 – 2022

Technical Skills

- **Languages:** Python, C, C++, JavaScript, SQL
- **Machine Learning / AI:** Machine Learning, Deep Learning, NLP, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Transformers, Attention Mechanisms, Hallucination Detection, Prompt Engineering, Agentic AI, Function Calling, PyTorch, Scikit-learn, Pandas, NumPy
- **Backend & Databases:** Flask, MySQL, Redis
- **DevOps & Cloud:** Docker, Kubernetes, Git, Vercel, CI/CD
- **Frontend:** React, Next.js, HTML, CSS, JavaScript, Tailwind CSS
- **Data & Visualization:** Pandas, Matplotlib, Power BI
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems

Research Experience

Research Intern — CDSAML Lab, PES University

Jan 2026 – Apr 2026

Under the guidance of Dr. S. Vanitha

- Co-developed CAR-HAL, an attention-based hallucination detection framework for RAG systems using a custom CAR metric and DeBERTa-v3.
- Evaluated on RGB/RAGTruth benchmarks with Mistral-7b-instruct-v0.2, identifying spurious faithfulness in 43.3% of queries.
- Optimized a hybrid FAISS and BM25 retrieval pipeline with GMM and logistic regression to improve classification reliability.

Publications

- CareNav: AR-Based Indoor Navigation and Multilingual Speech Translation for Hospitals — IEEE SmartNets 2026, Rome, Italy (presented July 2026).
- CAR-HAL: Attention-Based Hallucination Detection in RAG Systems — submitted to ICST 2026, IIT Patna.

Projects

CareNav — AR Hospital Navigation & Speech Translation App

Unity, ARCore, Flutter

- Built an augmented-reality indoor navigation system integrated with real-time multilingual speech translation to help hospital patients navigate complex layouts without separate tools.
- Achieved a 54.67% reduction in patient navigation time in evaluations conducted at PES Hospital.

Skylark Ops Coordinator (AeroOps) — AI Drone Operations *Python, Gemini 2.5 Flash, Google Sheets API*

Agent

- Designed a conversational AI agent to automate drone fleet, pilot roster, and mission-assignment coordination using an 11-tool function-calling schema built on Gemini 2.5 Flash.
- Used Google Sheets as a real-time operational database, enabling live scheduling updates and automated conflict detection.

Load-Balanced URL Shortener

Python, Redis, Docker, Kubernetes

- Developed a URL-shortening service using Redis for key-URL mapping, containerized with Docker and deployed across multiple replicas on Kubernetes.
- Implemented load balancing via Kubernetes Services and validated performance through stress testing.

Apache Kafka Clone

Python, SSL, Async Programming

- Built a secure, event-driven data-streaming system with a custom wire protocol and SSL encryption, simulating asynchronous producer-consumer workflows for distributed messaging.

Hospital Management System

MySQL

- Designed a 3NF-normalized relational schema for patients, staff, and resources, and optimized SQL queries to improve data retrieval efficiency.

Customer Behavior Data Analytics Pipeline

Python, SQL, Power BI

- Built an end-to-end analytics pipeline covering data cleaning, analysis, and dashboarding to support data-driven decision-making.

Certifications

- HackerRank — Problem Solving (Intermediate)
- Python for Data Science

Languages

English (Fluent) • Kannada (Native) • Hindi (Proficient)